

ZAOYANG, China

WMO: 572790

Lat: 32.1411N

Lon: 112.7553E

Elev: 417

StdP: 14.48

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.5	27.6	5.8	7.5	38.5	11.3	10.0	37.9	14.1	38.9	12.0	39.4	3.0	40	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	96.1	78.8	94.0	77.9	91.9	77.1	82.4	91.3	81.4	90.0	80.5	88.8	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.3	159.6	87.1	79.2	153.8	86.5	78.2	148.6	85.9	46.8	91.2	45.7	90.3	44.6	89.0	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.4	9.2	21.1	99.4	2.7	2.2	19.1	101.0	17.5	102.3	16.0	103.5	14.0	105.1
			19.4	84.1	2.7	1.3	17.4	85.0	15.8	85.8	14.2	86.5	12.2	87.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	38.1	43.1	53.1	63.6	72.8	79.8	83.4	81.7	74.1	64.6	52.6	42.0	
	DBStd	16.81	5.64	7.14	7.90	7.48	5.98	4.92	4.81	5.50	6.06	6.33	7.11	5.46	
	HDD50	941	368	210	57	4	0	0	0	0	0	0	50	252	
	HDD65	3119	832	612	377	112	9	0	0	0	6	86	373	712	
	CDD50	5508	1	18	153	411	707	893	1034	982	723	452	129	5	
	CDD65	2210	0	0	7	69	251	443	569	517	279	73	2	0	
	CDH74	22468	0	0	47	499	2086	4605	6788	5836	2186	408	13	0	
	CDH80	9471	0	0	6	123	745	1921	3207	2620	772	77	0	0	
Wind	WSAvg	4.1	3.7	4.0	4.6	4.7	4.6	4.6	4.6	4.0	3.7	3.4	3.6	3.5	
Precipitation	PrecAvg	33.9	0.6	0.9	1.9	3.3	3.9	4.0	6.5	5.2	3.5	2.3	1.4	0.5	
	PrecMax	49.4	1.7	3.5	4.6	16.3	10.6	10.5	12.6	13.7	13.9	6.4	5.1	1.8	
	PrecMin	21.6	0.0	0.0	0.1	0.3	0.8	0.4	1.3	1.2	0.2	0.0	0.1	0.0	
	PrecStd	7.2	0.4	0.7	1.1	3.0	1.8	2.6	3.3	3.5	2.7	1.5	1.1	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	67.4	79.7	88.1	93.6	96.8	99.2	98.9	95.3	87.0	75.9	62.0	
		MCWB	45.9	53.9	62.0	68.3	69.8	74.4	81.5	80.3	73.8	66.9	59.1	48.3	
	2%	DB	54.5	62.3	73.7	83.5	90.4	94.0	96.3	95.6	90.7	81.9	71.2	57.6	
		MCWB	43.8	49.9	59.0	66.5	69.0	74.4	80.6	80.0	73.6	64.1	56.6	46.4	
	5%	DB	51.0	58.5	69.5	79.8	87.2	91.6	94.2	92.9	87.5	78.9	67.3	54.2	
		MCWB	42.0	47.8	56.8	64.8	68.6	73.9	80.1	79.3	72.0	63.1	55.2	44.8	
	10%	DB	47.7	54.7	65.6	76.2	84.3	89.1	92.0	90.6	84.4	75.3	63.5	51.2	
		MCWB	40.3	45.7	54.6	62.8	68.0	73.9	79.6	78.5	70.8	62.3	53.9	43.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	56.2	65.3	72.6	76.2	81.6	84.2	83.6	80.1	70.8	62.0	51.2
			MCDB	55.7	63.4	75.4	81.4	85.2	90.6	93.1	91.4	90.8	78.6	69.6	56.9
2%		WB	45.4	52.0	61.5	69.5	73.8	79.7	82.7	82.3	77.9	67.9	59.4	48.6	
		MCDB	52.1	60.2	70.3	78.8	82.3	88.2	91.9	90.7	85.9	76.4	66.9	54.8	
5%		WB	43.1	49.3	58.4	67.2	72.2	78.2	81.8	81.3	75.8	66.1	57.3	46.5	
		MCDB	49.6	55.4	66.6	76.3	80.9	86.2	90.7	89.5	82.9	74.2	64.6	53.1	
10%		WB	40.9	47.2	55.7	64.8	70.5	76.8	80.9	80.3	73.6	64.3	55.2	44.3	
		MCDB	46.5	53.3	64.1	73.7	79.7	84.3	89.4	88.3	80.1	72.3	61.8	49.8	
Mean Daily Temperature Range		5% DB	MDBR	13.2	14.4	15.5	16.2	15.9	14.1	12.2	12.7	13.6	14.0	13.5	13.3
			MCDBR	19.6	21.2	20.9	21.7	21.0	17.8	15.0	15.6	17.8	19.5	19.3	19.2
	MCWBR		11.4	11.6	9.9	9.1	6.6	5.3	4.5	4.6	5.5	6.6	8.9	10.7	
	5% WB	MCDBR	17.4	17.7	19.1	18.7	15.6	13.8	13.6	13.5	15.0	15.5	16.0	16.6	
		MCWBR	11.0	10.6	10.1	9.5	7.0	5.4	5.3	5.1	6.6	6.9	8.4	10.0	
Clear-Sky Solar Irradiance	taub	0.642	0.702	0.703	0.696	0.687	0.733	0.702	0.734	0.722	0.709	0.680	0.631		
	taud	1.538	1.466	1.484	1.540	1.601	1.549	1.642	1.564	1.554	1.522	1.536	1.579		
	Ebn at Noon	173	180	195	207	211	200	206	196	189	175	162	166		
	Edh at Noon	74	87	91	89	84	89	81	86	84	81	73	67		
All-Sky Solar Radiation	RadAvg	673	792	1156	1373	1455	1485	1471	1376	1127	943	757	656		
	RadStd	101	174	108	138	137	145	189	178	152	131	109	112		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.58	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page